The Æther-flow Interpretation of Relativity

Next Structural Candidate: Non-Algebraic Momentum Completion Beyond the Exhausted Fixed-Route Compensator Class

Abstract

The fixed-route momentum-compensator exhaustion note closed the whole finite algebraic/gapped replacement program sharply. Keeping the bridge metric and the single-metric matter route fixed, every such completion collapses after elimination to one effective coefficient multiplying

$$\Psi^2, \qquad \Psi \equiv U^A \nabla_A S - \sigma_0,$$

and canceling that coefficient again annihilates the ordered-flow equation itself. The next live question is therefore no longer another algebraic/gapped redesign on the same route, but the first genuinely new structural candidate beyond that exhausted class.

This note writes that next candidate in the smallest honest fixed-route form presently available. Keep the tuned quartic exact-square block that already removes the old quartic observer correction. Keep the bridge metric

$$\mathfrak{g}_{AB} = G_{AB} - 2U_A U_B$$

and the single-metric matter route

$$S_{\text{matter}}[\Xi^* \mathfrak{g}, \psi].$$

Change only the remaining momentum block by adjoining one *non-algebraic* gapped scalar Z with elliptic spatial stiffness:

$$\mathcal{L}^{\text{na}}_{\Psi Z} = -\frac{1}{2} m_S^2 \Psi^2 + \alpha_Z Z \Psi - \frac{\beta_Z}{2} P^{AB} (\nabla_A Z)(\nabla_B Z) - \frac{1}{2} M_Z^2 Z^2, \qquad \alpha_Z \neq 0,\ \beta_Z > 0,\ M_Z^2 > 0.$$

The Z-equation is then

$$\mathcal{K}_Z Z = \alpha_Z \Psi, \qquad \mathcal{K}_Z \equiv M_Z^2 - \beta_Z \Delta_\perp,$$

so eliminating Z produces the operator-valued momentum kernel

$$\mathcal{K}_{\text{mom}} \equiv m_S^2 - \alpha_Z^2 \mathcal{K}_Z^{-1}$$

rather than one renormalized constant. In particular, the ordered-flow equation survives as the genuine differential relation

$$\nabla_A \big(U^A \mathcal{K}_{\text{mom}} \Psi \big) = 0$$

on the tuned quartic branch; unlike the exhausted algebraic/gapped class, the S-equation is not annihilated.

The note then performs the immediate retest demanded by the tracking layer. On the flat admissible background, after the same tuned quartic elimination and the same longitudinal

ordered-flow solve, the reduced scalar bridge channel is

$$\mathcal{L}^{(2)}_{\hat{\sigma},\mathrm{eff}} = -\frac{1}{2}\big(\partial_0\hat{\sigma} - \sigma_0\Phi_h\big)\Lambda_{\mathrm{na}}(-\nabla^2)\big(\partial_0\hat{\sigma} - \sigma_0\Phi_h\big),$$

with Fourier symbol

$$\Lambda_{\mathrm{na}}(k^2) = m_S^2 - \frac{\alpha_Z^2}{M_Z^2 + \beta_Z k^2}.$$

The corresponding flat-background observer-equation correction tensor therefore has leading projections

$$\rho_{\mathrm{na}} = \frac{1}{2}\big(\partial_0\hat{\sigma} - \sigma_0\Phi_h\big)\Lambda_{\mathrm{na}}(-\nabla^2)\big(\partial_0\hat{\sigma} - \sigma_0\Phi_h\big),$$

$$J_i^{\mathrm{na}} = -\Lambda_{\mathrm{na}}(-\nabla^2)\big(\partial_0\hat{\sigma} - \sigma_0\Phi_h\big)\partial_i\hat{\sigma},$$

with every remaining spatial component controlled by the same nonconstant symbol $\Lambda_{\mathrm{na}}(k^2)$.

That is the first honest post-exhaustion verdict. The new branch is genuinely beyond the exhausted finite algebraic/gapped class because it keeps a real ordered-flow equation. But the generic observer remainder still vanishes only if $\Lambda_{\mathrm{na}}(k^2) \equiv 0$ for all k, which forces $\beta_Z = 0$ and hence collapses straight back to the exhausted algebraic route. The smallest non-algebraic momentum completion is therefore architecturally honest but coefficient-level negative. The note stops there, without reopening reduced-system, local-coercive, or theorem work. At fixed bridge metric and single-metric matter route, the next admissible structural direction is now the explicit symmetry/quotient route; failing that, a later reconsideration of the bridge metric or of the matter route becomes the next larger move.

1. Introduction

The previous note closed a whole mechanism class rather than a single branch. Once the full finite algebraic/gapped momentum-compensator family was written down, every least-drift enlargement of the old single-Z replacement mechanism reduced to one effective coefficient multiplying Ψ^2. Canceling the generic observer remainder at flat-background coefficient level again meant canceling that last coefficient, and that again removed the ordered-flow equation itself. That fixed-route class is now exhausted.

The next manuscript burden is therefore more structural than the last one. It is not another reduced-system note on the failed branch. It is not another theorem note on the same closed route. It is not another algebraic/gapped compensator redesign disguised by extra notation. The next burden is to choose the first candidate whose momentum completion is genuinely *not* reducible to one compensator-renormalized Ψ^2 coefficient.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays within that boundary. It does not claim that every non-algebraic momentum completion is now ruled out in principle. It does something narrower and more disciplined. It selects the first honest post-exhaustion fixed-route candidate, proves immediately that the ordered-flow equation survives as a genuine equation on that branch, retests the generic observer remainder $8\pi G_N T^{\mathrm{St}}_{\mu\nu}$ at flat-background quadratic level by computing the explicit observer-equation correction tensor, and stops there before any reduced-system or theorem work is reopened.

2. What the Exhaustion Verdict Leaves Open

Definition 1 (Honest post-exhaustion fixed-route candidate)**.** *An* honest post-exhaustion fixed-route candidate *is a replacement bridge/completion branch that:*

1. *keeps the bridge metric $\mathfrak{g}_{AB} = G_{AB} - 2U_A U_B$ and the single-metric matter route fixed;*
2. *is not reducible, after elimination of its new compensator variables, to one constant renormalization of Ψ^2;*
3. *leaves a genuine ordered-flow equation for S rather than reproducing the identity $0 = 0$;*
4. *is carried immediately through flat-background quadratic reduction and the explicit observer-equation correction tensor before any reduced-system, local-coercive, or broader theorem step is attempted.*

Proposition 1 (First post-exhaustion choice)**.** *Among the two fixed-route directions still left open by the exhaustion verdict, the first one that should be tested is a non-algebraic momentum completion rather than a symmetry/quotient principle.*

Proof. The exhaustion verdict closed only the finite algebraic/gapped compensator route. The smallest remaining structural move that still keeps the bridge metric and matter route fixed is therefore to change the momentum block itself while retaining the action-based branch logic. An explicit symmetry or quotient principle is a larger conceptual move because it changes the physical status of the ordered-flow label rather than first testing whether a genuine ordered-flow equation can survive on a minimally altered fixed route. Hence the non-algebraic completion is the first admissible direction to test. □

Remark 1. *Proposition 1 is only an ordering statement. If the first non-algebraic route fails honestly, the symmetry/quotient route remains open as the next fixed-route direction.*

3. Non-Algebraic Momentum Completion Candidate

Definition 2 (Elliptic non-algebraic momentum completion)**.** *The* elliptic non-algebraic momentum completion *is the action*

$$S_{\mathrm{na}} = \frac{c^3}{16\pi G_N} \int_{\mathcal{A}} d^4X \sqrt{G}\,(R[\mathfrak{g}] - 2\Lambda_{\mathcal{A}}) + \int_{\mathcal{A}} d^4X \sqrt{G}\,\mathcal{L}_{\mathrm{aux,na}} + S_{\mathrm{matter}}[\Xi^*\mathfrak{g}, \psi], \tag{1}$$

with bridge metric

$$\mathfrak{g}_{AB} = G_{AB} - 2U_A U_B, \qquad P_{AB} = G_{AB} - U_A U_B, \tag{2}$$

and auxiliary Lagrangian

$$\begin{aligned}\mathcal{L}_{\text{aux,na}} = \lambda_U(G_{AB}U^AU^B - 1) - \frac{1}{2}\left(\frac{\sqrt{\lambda_4}}{M_*}\Delta_\perp S - M_Y Y\right)^2 \\ - \frac{1}{2}m_S^2\Psi^2 + \alpha_Z Z\Psi - \frac{\beta_Z}{2}P^{AB}(\nabla_A Z)(\nabla_B Z) - \frac{1}{2}M_Z^2 Z^2 \\ - \frac{1}{2}\delta_{IJ}P^{AB}(D_A Q^I)(D_B Q^J) - \frac{1}{2}\widehat{M}_{IJ}Q^I Q^J - \frac{\kappa_\theta}{2}(\nabla_A U^A)^2 - \frac{\kappa_\Omega}{4}\Omega_{AB}\Omega^{AB},\end{aligned} \tag{3}$$

where

$$\Psi \equiv U^A\nabla_A S - \sigma_0, \qquad \Delta_\perp S \equiv \nabla_A(P^{AB}\nabla_B S), \qquad \Omega_{AB} \equiv P_A{}^C P_B{}^D \nabla_{[C}U_{D]}, \tag{4}$$

and

$$m_S^2 > 0, \qquad \lambda_4 > 0, \qquad M_Y^2 > 0, \qquad M_Z^2 > 0, \qquad \beta_Z > 0, \qquad \alpha_Z \neq 0, \qquad \widehat{M}_{IJ} > 0, \qquad \kappa_\theta > 0, \qquad \kappa_\Omega > 0. \tag{5}$$

Remark 2. *Definition 2 keeps the tuned quartic exact-square block, the bridge metric, and the single-metric matter route fixed. The only new structure is that the remaining momentum compensator is no longer algebraic: the new field Z is elliptic and gapped rather than pointwise solvable.*

Proposition 2 (Differential Z-equation and non-algebraic momentum kernel)**.** *The Euler–Lagrange equation for Z is*

$$\mathcal{K}_Z Z = \alpha_Z\Psi, \qquad \mathcal{K}_Z \equiv M_Z^2 - \beta_Z\Delta_\perp. \tag{6}$$

On the decaying branch $\mathcal{K}_Z$ is positive elliptic and invertible. After solving (6) *for Z, the momentum block becomes*

$$\mathcal{L}_{\Psi,\text{red}} = -\frac{1}{2}\,\Psi\,\mathcal{K}_{\text{mom}}\,\Psi, \qquad \mathcal{K}_{\text{mom}} \equiv m_S^2 - \alpha_Z^2\mathcal{K}_Z^{-1}. \tag{7}$$

Proof. Varying (3) with respect to Z gives

$$\beta_Z\Delta_\perp Z - M_Z^2 Z + \alpha_Z\Psi = 0,$$

which is (6). Since $M_Z^2 > 0$ and $\beta_Z > 0$, the operator $\mathcal{K}_Z$ is positive elliptic on the decaying branch and therefore invertible. Substituting

$$Z = \alpha_Z\mathcal{K}_Z^{-1}\Psi$$

back into the momentum block yields

$$\alpha_Z Z\Psi - \frac{1}{2}Z\mathcal{K}_Z Z = \frac{\alpha_Z^2}{2}\Psi\mathcal{K}_Z^{-1}\Psi,$$

which combines with $-\frac{1}{2}m_S^2\Psi^2$ to give (7). □

Corollary 1 (Why this candidate escapes the exhausted class)**.** *The branch of Definition 2 is not reducible to one renormalized constant multiplying Ψ^2.*

Proof. On the flat admissible background the Fourier symbol of $\mathcal{K}_{\text{mom}}$ is

$$\Lambda_{\text{na}}(k^2) = m_S^2 - \frac{\alpha_Z^2}{M_Z^2 + \beta_Z k^2}. \tag{8}$$

Because $\beta_Z > 0$, $\Lambda_{\rm na}(k^2)$ depends nontrivially on k^2 and is therefore not one constant renormalization of Ψ^2. $\square$

4. Survival of the Ordered-Flow Equation

Definition 3 (Non-algebraic quartic-tuned branch)**.** *The* non-algebraic quartic-tuned branch *of Definition 2 is the branch on which the quartic exact-square auxiliary satisfies*

$$\mathcal{A}_Y \equiv \frac{\sqrt{\lambda_4}}{M_*}\Delta_\perp S - M_Y Y = 0. \tag{9}$$

Proposition 3 (Ordered-flow equation on the non-algebraic branch)**.** *The S-equation of Definition 2 is*

$$\frac{\sqrt{\lambda_4}}{M_*}\Delta_\perp \mathcal{A}_Y + \nabla_A\Big[U^A\big(m_S^2\Psi - \alpha_Z Z\big)\Big] = 0. \tag{10}$$

On the non-algebraic quartic-tuned branch of Definition 3, this reduces to

$$\nabla_A\Big(U^A\mathcal{K}_{\rm mom}\Psi\Big) = 0. \tag{11}$$

Proof. Varying (3) with respect to S gives one contribution from the quartic exact square and one from the momentum block. Integration by parts yields (10). On the tuned branch $\mathcal{A}_Y = 0$, and substitution of $Z = \alpha_Z\mathcal{K}_Z^{-1}\Psi$ from Proposition 2 gives

$$m_S^2\Psi - \alpha_Z Z = \Big(m_S^2 - \alpha_Z^2\mathcal{K}_Z^{-1}\Big)\,\Psi = \mathcal{K}_{\rm mom}\Psi,$$

which is (11). $\square$

Theorem 1 (The ordered-flow equation survives genuinely)**.** *On the honest non-algebraic branch $\beta_Z > 0$, the ordered-flow equation* (11) *is a genuine equation and does not collapse to the identity $0 = 0$.*

Proof. If (11) collapsed identically, then $\mathcal{K}_{\rm mom}$ would vanish as an operator. But its flat-background symbol is $\Lambda_{\rm na}(k^2)$ from (8). Vanishing of that symbol for all k would require

$$m_S^2 = \frac{\alpha_Z^2}{M_Z^2 + \beta_Z k^2}$$

for every k, impossible when $\beta_Z > 0$. Therefore $\mathcal{K}_{\rm mom}$ is not the zero operator, and (11) is a genuine differential equation for the ordered-flow sector. $\square$

Remark 3. *Theorem 1 is precisely the structural point that the exhausted algebraic/gapped class could not achieve. The present candidate may still fail elsewhere, but it does* not *fail by annihilating the ordered-flow equation.*

5. Flat-Background Quadratic Reduction

Use the same admissible homogeneous flat background as in the tuned compensator and fixed-route exhaustion notes:

$$\bar{G}_{AB} = \delta_{AB}, \qquad \bar{U}^A = \delta^A{}_0, \qquad \bar{S} = \sigma_0 X^0, \qquad \bar{Q}^I = 0, \qquad \bar{Y} = 0, \qquad \bar{Z} = 0. \tag{12}$$

Write perturbations as

$$G_{AB} = \delta_{AB} + H_{AB}, \qquad U^A = \bar{U}^A + V^A, \qquad S = \sigma_0 X^0 + \sigma, \qquad Y = y, \qquad Z = z, \tag{13}$$

with scalar metric variables

$$h_{00} = -2\phi = -H_{00}, \qquad h_{0i} = S_i + \partial_i B = -H_{0i} - 2V_i, \qquad V_i = V_i^T + \partial_i \chi. \tag{14}$$

Proposition 4 (Linearized scalar blocks)**.** *On the background* (12)*,*

$$\Psi^{(1)} = \partial_0 \sigma - \sigma_0 \phi, \tag{15}$$

$$\left(\Delta_\perp S\right)^{(1)} = \nabla^2 \sigma + \sigma_0 \nabla^2 (B + \chi). \tag{16}$$

Proof. These are the same linearized scalar blocks already derived in the tuned compensator and fixed-route exhaustion notes. The present branch changes only the remaining momentum-completion block. □

Proposition 5 (Quadratic nonmetric scalar action)**.** *At fixed bridge perturbation $h_{\mu\nu}$, the complete flat-background quadratic nonmetric scalar action of Definition 2 is*

$$\begin{aligned}\mathcal{L}^{(2)}_{\text{aux,na}} = & -\frac{1}{2} q^I \Big((-\nabla^2)\delta_{IJ} + \widehat{M}_{IJ} \Big) q^J - \frac{\kappa_\theta}{2} \left(\nabla^2 \chi + \frac{1}{2} \partial_0 H \right)^2 \\ & - \frac{1}{2} \left(\frac{\sqrt{\lambda_4}}{M_*} \Xi - M_Y y \right)^2 - \frac{1}{2} m_S^2 \Psi_h^2 + \alpha_Z z \, \Psi_h \\ & - \frac{\beta_Z}{2} \partial_i z \, \partial^i z - \frac{1}{2} M_Z^2 z^2,\end{aligned} \tag{17}$$

where

$$\Psi_h \equiv \partial_0 \sigma - \sigma_0 \phi, \qquad \Xi \equiv \nabla^2 \sigma + \sigma_0 \nabla^2 (B + \chi). \tag{18}$$

Proof. Expand Definition 2 to quadratic order around (12) using Proposition 4. Relative to the fixed-route exhaustion note, the only change is that the algebraic z^2 mass term is replaced by the elliptic block $(\beta_Z/2)\partial_i z \partial^i z + (M_Z^2/2) z^2$. □

Proposition 6 (Elliptic z-solve and reduced scalar bridge channel)**.** *The linearized z-equation is*

$$\left(M_Z^2 - \beta_Z \nabla^2 \right) z = \alpha_Z \Psi_h. \tag{19}$$

After solving (19)*, imposing the tuned quartic branch $\mathcal{A}_Y = 0$, and eliminating χ on the decaying*

branch, the reduced scalar bridge channel is

$$\mathcal{L}^{(2)}_{\hat{\sigma},\mathrm{eff}} = -\frac{1}{2}(\partial_0\hat{\sigma} - \sigma_0\Phi_h)\Lambda_{\mathrm{na}}(-\nabla^2)(\partial_0\hat{\sigma} - \sigma_0\Phi_h), \tag{20}$$

where

$$\Lambda_{\mathrm{na}}(-\nabla^2) \equiv m_S^2 - \alpha_Z^2\left(M_Z^2 - \beta_Z\nabla^2\right)^{-1}, \tag{21}$$

and

$$\Phi_h \equiv \phi + \partial_0\nabla^{-2}\Psi_h^{\mathrm{long}}, \qquad \hat{\sigma} \equiv \sigma + \sigma_0\nabla^{-2}\Psi_h^{\mathrm{long}}, \qquad \Psi_h^{\mathrm{long}} \equiv \nabla^2 B - \frac{1}{2}\partial_0 H. \tag{22}$$

Proof. Equation (19) follows from varying (17) with respect to z. Substituting

$$z = \alpha_Z\left(M_Z^2 - \beta_Z\nabla^2\right)^{-1}\Psi_h$$

into (17) produces the non-algebraic momentum kernel (21). On the tuned quartic branch, the full y-block vanishes after the algebraic solve

$$M_Y y = \frac{\sqrt{\lambda_4}}{M_*}\Xi.$$

The remaining χ-equation is therefore the same tuned one already recorded:

$$\nabla^2\chi = -\frac{1}{2}\partial_0 H.$$

Hence $\Psi_h^{\mathrm{long}} = \nabla^2(B+\chi)$, and the redefinitions (22) give

$$\partial_0\hat{\sigma} - \sigma_0\Phi_h = \partial_0\sigma - \sigma_0\phi = \Psi_h.$$

Substitution yields (20). □

Remark 4. *At the level of the scalar bridge channel, Proposition 6 is the first explicit sign that the present candidate really leaves the exhausted class. The whole momentum sector is now controlled by the operator-valued kernel $\Lambda_{\mathrm{na}}(-\nabla^2)$, not by one renormalized coefficient.*

Definition 4 (Fourier symbol of the reduced momentum kernel)**.** *For a spatial Fourier mode of wave number $k = |\mathbf{k}|$, the symbol of $\Lambda_{\mathrm{na}}(-\nabla^2)$ is*

$$\Lambda_{\mathrm{na}}(k^2) = m_S^2 - \frac{\alpha_Z^2}{M_Z^2 + \beta_Z k^2}. \tag{23}$$

6. Explicit Observer-Equation Correction Tensor

Remark 5. *On the pure momentum channel* (20)*, the metric self-energy is no longer the right stopping test by itself. For nonzero temporal frequency, integrating out $\hat{\sigma}$ imposes*

$$\omega(\omega\hat{\sigma} - \sigma_0\Phi_h) = 0$$

and therefore gives a vanishing quadratic metric self-energy. That is exactly why the tracking layer now requires the explicit observer-equation correction tensor rather than self-energy alone.

Definition 5 (Non-algebraic flat-background correction tensor)**.** *At the flat-background quadratic scope, define the* non-algebraic correction tensor $T^{\rm na}_{\mu\nu}$ *by*

$$T^{\rm na}_{\mu\nu} = \rho_{\rm na}\, n_\mu n_\nu + 2n_{(\mu} J^{\rm na}_{\nu)} + S^{\rm na}_{\mu\nu}, \tag{24}$$

where $J^{\rm na}_\mu n^\mu = 0$, $S^{\rm na}_{\mu\nu} n^\nu = 0$, *and the leading source projections of* (20) *are*

$$\rho_{\rm na} = \frac{1}{2}(\partial_0\hat{\sigma} - \sigma_0\Phi_h)\Lambda_{\rm na}(-\nabla^2)(\partial_0\hat{\sigma} - \sigma_0\Phi_h), \tag{25}$$

$$J^{\rm na}_i = -\Lambda_{\rm na}(-\nabla^2)(\partial_0\hat{\sigma} - \sigma_0\Phi_h)\partial_i\hat{\sigma}, \tag{26}$$

with every remaining spatial component of $S^{\rm na}_{\mu\nu}$ *determined by the same nonconstant symbol* $\Lambda_{\rm na}(k^2)$. *The corresponding observer-equation correction tensor is*

$$\mathcal{C}^{\rm na,br}_{\mu\nu} = 8\pi G_N\, T^{\rm na}_{\mu\nu}. \tag{27}$$

Theorem 2 (Flat-background verdict on the first non-algebraic candidate)**.** *For the non-algebraic candidate of Definition 2, the following are equivalent:*

1. *the flat-background observer-equation correction tensor* $\mathcal{C}^{\rm na,br}_{\mu\nu}$ *vanishes identically for generic modes;*
2. $\Lambda_{\rm na}(k^2) \equiv 0$ *for all* k*;*
3. $\beta_Z = 0$ *and* $M_Z^2 = \alpha_Z^2/m_S^2$.

Hence on the honest non-algebraic branch $\beta_Z > 0$, *the generic observer remainder survives at flat-background quadratic level.*

Proof. By Definition 5, every displayed component of $T^{\rm na}_{\mu\nu}$ is controlled by $\Lambda_{\rm na}(-\nabla^2)$, and every remaining spatial component carries the same Fourier symbol $\Lambda_{\rm na}(k^2)$. Therefore item (2) implies item (1). Conversely, if $\mathcal{C}^{\rm na,br}_{\mu\nu} \equiv 0$, then in particular $\rho_{\rm na} \equiv 0$. For a generic Fourier mode this forces $\Lambda_{\rm na}(k^2) = 0$, so item (1) implies item (2).

Now suppose item (2) holds. Then

$$m_S^2 = \frac{\alpha_Z^2}{M_Z^2 + \beta_Z k^2}$$

for all k. The right-hand side is independent of k only if $\beta_Z = 0$, and then one must also have $M_Z^2 = \alpha_Z^2/m_S^2$. Hence item (2) implies item (3). Finally, item (3) makes $\Lambda_{\rm na}(k^2) \equiv 0$, so item (3) implies item (2). This proves the equivalence. The last sentence follows immediately because the honest non-algebraic branch assumes $\beta_Z > 0$. □

Corollary 2 (Status of the first post-exhaustion candidate)**.** *The branch of Definition 2 is the first genuinely new structural candidate beyond the exhausted algebraic/gapped fixed route, but it is coefficient-level negative as a generic observer-remainder removal mechanism.*

Proof. Corollary 1 shows that the branch is genuinely beyond the exhausted class, while Theorem 1 shows that its ordered-flow equation survives honestly. Theorem 2 then shows that its explicit observer-equation correction tensor cannot vanish generically without collapsing back to the algebraic class. Therefore the branch is structurally honest but coefficient-level negative. □

Corollary 3 (What the next manuscript should be)**.** *No reduced-system, local-coercive, or theorem work should be reopened on the present candidate. If the bridge metric and single-metric matter route are still to be preserved, the next fixed-route manuscript should instead test an explicit symmetry or quotient principle making the ordered-flow label physically redundant. If that cannot be made honest, a later reconsideration of the bridge metric or of the single-metric matter route becomes the next larger move.*

Proof. By Corollary 2, the present candidate has already failed the coefficient-level observer-remainder test that must be passed before broader nonlinear work is reopened. The only fixed-route direction not yet tested in order is therefore the explicit symmetry/quotient route. If that also fails to produce an honest branch, the next remaining move is the larger structural reconsideration of the bridge metric or the matter route. □

7. Conclusion

The present manuscript establishes four points.

1. The first honest fixed-route candidate beyond the exhausted finite algebraic/gapped momentum-compensator class is a non-algebraic momentum completion, not another algebraic/gapped redesign.

2. On that candidate, the ordered-flow equation survives as a genuine equation; unlike the exhausted class, the S-equation is not annihilated.

3. At flat-background quadratic level the reduced scalar bridge channel is governed by the nonconstant kernel $\Lambda_{\mathrm{na}}(k^2) = m_S^2 - \alpha_Z^2/(M_Z^2 + \beta_Z k^2)$, and the explicit observer-equation correction tensor is controlled by that same symbol.

4. The generic observer remainder can vanish only by forcing $\beta_Z = 0$, which collapses the candidate back to the already exhausted algebraic route.

This is therefore the next honest stopping point. The first genuinely new structural candidate has now been written and tested at the required flat-background quadratic observer-tensor level. It preserves the bridge metric, preserves the single-metric matter route, and preserves a genuine ordered-flow equation, but it does not remove the generic observer remainder. The next fixed-route structural burden is now the explicit symmetry/quotient route, and no reduced-system or theorem work should be reopened before that decision is made.